

Conditional Face Attribute Generation and Classification

Final Group Project Report

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Group 3: Yuan (Irene) Xia, Zachary Cardell, Jiatong Peng

1. Introduction

This project develops a complete deep learning pipeline for attribute-controlled face generation and evaluation using the CelebA dataset, which contains 202,599 celebrity face images each annotated with 40 binary facial attributes. The system addresses two complementary problems: (1) classifying all 40 attributes from a real or generated face image, and (2) synthesizing new face images conditioned on a user-specified attribute vector.

The pipeline consists of three integrated components. First, a multi-label attribute classifier based on DenseNet169 and a Bidirectional LSTM (Irene Xia) that predicts all 40 attributes simultaneously from a 224×224 face image. Second, a Conditional Variational Autoencoder or cVAE (Zachary Cardell) that generates 128×128 RGB face images conditioned on a 40-dimensional binary attribute vector. Third, an interactive Streamlit web demo (Jiatong Peng) that integrates both models, enabling users to select attributes, generate faces, and immediately see classifier-based attribute evaluation.

The end-to-end workflow is: (1) train the classifier on real CelebA images; (2) train the cVAE on the same training split with attribute conditioning; (3) generate synthetic images and pass them through the classifier to measure attribute fidelity; (4) expose this pipeline interactively in the demo with conflict detection and real-time evaluation.

Component	Developer	Description
Multi-label Classifier (train script.py)	Irene Xia	DenseNet169 + BiLSTM for all 40 attributes; CLAHE preprocessing, disk caching, pos_weight, SWA, per-class threshold tuning
cVAE (cVAE_train_option1_lpips.py, evaluate_generated_full.py)	Zachary Cardell	128×128 conditional VAE with LPIPS perceptual loss, KL warm-up, 6-variant ablation, full 10k evaluation pipeline
Streamlit Demo (streamlit_demo.py)	Jiatong Peng	Web interface with attribute selection, hard/soft conflict detection, generation, classifier evaluation, and image upload

Table 1. Division of work across the group.

This report is organized as follows: Section 2 describes the CelebA dataset; Section 3

details both deep learning architectures; Section 4 explains the experimental setup; Section 5 presents results from all three components; Section 6 concludes with lessons learned and future directions.

2. Description of the Dataset

2.1 CelebA Overview

The Large-scale CelebFaces Attributes (CelebA) dataset (Liu et al., 2015) contains 202,599 face images of celebrities collected from the internet. Each image is annotated with 40 binary attributes covering hair color, facial structure, makeup, accessories, and demographic characteristics. The dataset is officially partitioned into 162,770 training, 19,867 validation, and 19,962 test images, maintaining celebrity-level separation across splits to prevent data leakage.

2.2 Class Imbalance

A critical characteristic of CelebA is severe class imbalance across attributes. Positive-example rates range from 83.2% (No_Beard) to 2.1% (Bald), spanning nearly two orders of magnitude. This imbalance directly motivates the use of weighted loss functions in the classifier. Table 2 illustrates the range.

Attribute	Positive Rate (%)	Category	Challenge
No_Beard	83.2	Facial Hair	Common — easy to learn
Young	77.3	Age	Common but somewhat subjective
Smiling	48.2	Expression	Balanced
Eyeglasses	6.5	Accessories	Rare but visually distinctive
Wearing_Necktie	7.0	Accessories	Rare and often occluded
Bald	2.1	Hair	Very rare; strong visual signal

Table 2. Positive-example rates for selected CelebA attributes.

2.3 Preprocessing

Both models share a common preprocessing philosophy but differ in output resolution. For the classifier, images are processed at 224×224; for the cVAE, at 128×128. The shared steps are:

- CLAHE (Contrast Limited Adaptive Histogram Equalization) applied in LAB color space (clipLimit=2.0, tileGridSize=8×8) to enhance local contrast while

preserving color distribution — critical for color-dependent attributes such as Blond_Hair and Black_Hair.

- Gaussian blur (3×3 kernel) to reduce high-frequency noise.
- Resize and normalization using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) for the classifier; normalization to $[-1, 1]$ for the cVAE decoder's tanh output.

For the classifier, processed tensors are saved to disk after the first pass using GPU-safe multiprocessing (spawn context, not fork), eliminating 3.25M redundant preprocessing operations across 20 epochs. For the cVAE, images are center-cropped before resizing.

3. Description of the Deep Learning Networks

3.1 Multi-Label Classifier: DenseNet169 + BiLSTM

Motivation

Multi-label classification requires independent binary predictions for each of the 40 attributes. A standard global-average-pooling head discards spatial ordering, but face images have a natural top-to-bottom structure (hair → forehead → eyes → nose → mouth → chin). We designed a CNN-RNN hybrid that treats the spatial rows of the feature map as a temporal sequence.

Architecture

The DenseNet169 backbone (Huang et al., 2017), pretrained on ImageNet, processes each 224×224 image through four dense blocks with growth rate $k=32$, producing a (B, 1664, 7, 7) feature map. Dense connectivity creates $L(L+1)/2$ connections for L layers:

$$x_l = H_l([x_0, x_1, \dots, x_{l-1}])$$

The feature map is reshaped into a sequence of 7 row vectors, each of dimension 11,648 (7 columns × 1664 channels), and processed by a two-layer Bidirectional LSTM (hidden size 512):

$$h_{final} = \text{concat}(h_{fwd_last}, h_{bwd_last}) \in \mathbb{R}^{1024}$$

A classification head applies $\text{Linear}(1024 \rightarrow 256) \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(256 \rightarrow 40)$ to produce 40 independent logits.

Loss Function

BCEWithLogitsLoss with per-class positive weights addresses class imbalance:

$$\text{pos_weight}_c = (N - N_{\text{pos}_c}) / (N_{\text{pos}_c} + \epsilon)$$

For Bald (2.1% positive), $\text{pos_weight} \approx 46$, penalizing false negatives 46× more than false positives. This prevents the model from defaulting to always predicting negative for rare attributes.

3.2 Conditional VAE (cVAE)

Motivation

A Conditional VAE (Sohn et al., 2015) extends the standard VAE so that both encoder and decoder are conditioned on an auxiliary variable — here the 40-dimensional binary CelebA attribute vector c . At generation time, choosing c controls which attributes appear in the decoded image, while the latent z controls residual within-class variation (identity, lighting, pose).

Architecture

The encoder is a five-layer convolutional stack (3→64→128→256→512→512 channels, LeakyReLU activations, BatchNorm) that processes 128×128 RGB images. The final feature map is flattened, concatenated with a 128-dim attribute embedding $e(c)$, and projected to μ and $\log\sigma^2$ for a 256-dimensional latent:

$$q_{\phi}(z|x, c) = N(\mu_{\phi}(x, c), \sigma^2_{\phi}(x, c))$$

Sampling uses the reparameterization trick: $z = \mu + \sigma \odot \epsilon$, $\epsilon \sim N(0, I)$. The decoder mirrors the encoder with five transposed-convolutional blocks (512→512→256→128→64→3), concatenating z with the same $e(c)$ to reconstruct the image via tanh output. Total parameters: approximately 21 million.

Loss Function

The training objective combines three terms:

$$L = MSE(x, \hat{x}) + 1000 \cdot LPIPS(x, \hat{x}) + \beta(t) \cdot KL(q(z|x, c) \parallel N(0, I))$$

LPIPS (Learned Perceptual Image Patch Similarity) compares deep features from a pretrained AlexNet rather than raw pixels, recovering sharper reconstructions that pixel-wise MSE alone cannot produce. The weight of 1000 was chosen empirically. The KL weight $\beta(t)$ is linearly annealed from 0 at epoch 0 to 1.0 at epoch 10 to prevent posterior collapse — a failure mode where the encoder drives KL to zero and the decoder ignores z .

3.3 Streamlit Demo

The interactive demo integrates both models with three functional pages: (1) attribute selection with real-time conflict detection, (2) face generation with classifier evaluation,

and (3) image upload with direct attribute classification. The demo runs the complete pipeline — cVAE generation → preprocessing → classifier inference → per-attribute probability display — in a single user interaction.

In addition to basic integration, the Streamlit demo includes several system-level design features to improve usability and robustness.

First, a conflict detection mechanism was implemented to identify both hard conflicts (e.g., Bald + Bangs) and soft conflicts (e.g., Female + Mustache). These conflicts are detected in real time and displayed as warnings or errors in the interface.

Second, a model selection module allows users to dynamically switch between different trained cVAE models. Since different models may use different architectures, a unified loading function was implemented to select the correct model class based on the model type.

Third, an image upload and classification module was added. This allows users to upload external images and evaluate them using the classifier. A face detection step is included to prevent invalid inputs and ensure the classifier is only applied to face images.

These features extend the demo beyond a simple visualization tool and make it a flexible platform for analyzing model behavior.

4. Experimental Setup

4.1 Data Splits

Both models use the official CelebA train/val/test partition (162,770/19,867/19,962). The validation set is used for hyperparameter tuning and threshold selection; the test set is held out for final evaluation only. Classifier thresholds are tuned on the validation set and never on the test set to prevent data leakage.

4.2 Classifier Training Configuration

Hyperparameter	Value	Rationale
Image size	224 × 224	Standard ImageNet resolution for DenseNet169
Batch size	32	Fits 8 GB GPU; stable gradient estimates
Learning rate	1e-4	Standard fine-tuning rate; prevents catastrophic forgetting
Optimizer	AdamW (wd=1e-2)	Fixes Adam’s broken weight decay
Scheduler	CosineAnnealingWarmRestarts (T_0=5)	Periodic restarts to escape local minima
Max epochs	20 (early stopping patience=8)	SWA needs epochs 8+ to accumulate averages
SWA	From epoch 8, swa_lr=5e-5	Weight averaging for flatter loss basin
Gradient clipping	max_norm=1.0	Prevents explosion from large pos_weight values
AMP	autocast + GradScaler	~2× throughput with float16 forward pass

Table 3. Classifier training hyperparameters.

4.3 cVAE Training Configuration

Hyperparameter	Value	Rationale
Image size	128 × 128	4× more pixels than 64×64; better attribute detail
Batch size	128	Larger batch stable for VAE KL estimation
Learning rate	1e-4 (Adam)	Standard VAE training rate

Latent dimension	256	Balance between capacity and regularization
KL warm-up	0 $\rightarrow$ 1.0 over 10 epochs	Prevents posterior collapse in early training
LPIPS weight	1000	Dominant reconstruction signal for sharper faces
Epochs	20	Final checkpoint used (not best val loss)

Table 4. cVAE training hyperparameters.

4.4 Evaluation Metrics

The classifier is evaluated on macro-averaged F1, micro F1, and Hamming loss on the 19,962-image test set. Per-class F1 reveals which attributes are hardest to classify. For the cVAE, we generate 10,000 images conditioned on real validation-set attribute vectors, run them through the classifier with Irene’s per-attribute thresholds, and compute per-attribute F1 against the target conditioning vector. The gap (Δ) between generated-image F1 and real-image F1 isolates conditioning fidelity from raw classifier accuracy. FID measures distributional image quality.

4.5 Per-Class Threshold Tuning

For each of the 40 attributes, we sweep decision thresholds from 0.1 to 0.9 (step 0.05) on the validation set, selecting the value that maximizes per-class F1. A fixed 0.5 threshold fails for rare attributes — for Bald, the model outputs probabilities around 0.1–0.3 even for true positives, so 0.5 would classify nearly all images as negative. These tuned thresholds are then applied identically to both the classifier test evaluation and the cVAE evaluation pipeline.

5. Results

5.1 Classifier Performance on Real Images

Table 5 reports overall classifier performance on the held-out test set of 19,962 images.

Metric	Value	Interpretation
Macro F1 (test)	0.7413	Equal weight to all 40 attributes; rare attributes included
Micro F1 (test)	0.8016	Frequency-weighted; common attributes dominate
Hamming Loss (test)	0.0964	~9.6% of attribute-image pairs mispredicted; ~3.9 errors per image

Table 5. Overall classifier performance on the CelebA test set (19,962 images).

The gap between micro F1 (0.8016) and macro F1 (0.7413) reflects class imbalance: common attributes (No_Beard: F1=0.978, Young: F1=0.924) dominate the micro average, while rare subjective attributes drag down the macro. SWA contributed +0.012 val F1 improvement at no additional training cost.

5.2 Per-Attribute Classifier Results

Table 6 reports per-attribute F1, precision, and recall ranked by F1 score. Performance spans from 0.979 (Male) to 0.465 (Narrow_Eyes), a range of 0.514 points.

Rank	Attribute	F1	Precision	Recall	N Positive
1	Male	0.9788	0.9770	0.9806	7,715
2	No_Beard	0.9784	0.9725	0.9844	17,041
3	Eyeglasses	0.9779	0.9753	0.9806	1,289
4	Mouth_Slightly_Open	0.9363	0.9410	0.9317	9,883
5	Wearing_Lipstick	0.9284	0.9664	0.8933	10,418
⋮	⋮	⋮	⋮	⋮	⋮
36	Double_Chin	0.5794	0.4938	0.7010	913
37	Oval_Face	0.5654	0.4632	0.7255	5,901
38	Wearing_Necklace	0.5228	0.4885	0.5623	2,753

39	Big_Lips	0.5084	0.5941	0.4442	6,528
40	Narrow_Eyes	0.4654	0.6488	0.3629	2,968

Table 6. Per-attribute classifier results on test set (selected rows).

5.3 Key Finding: Visual Distinctiveness Over Sample Count

A critical observation is that sample count does not predict F1. Eyeglasses achieves $F1=0.978$ despite only 1,289 positive examples (6.5%) because glasses produce an unambiguous visual signal. Narrow_Eyes (2,968 positives) scores only 0.465 because the difference is subtle and annotations are inconsistent. Visual distinctiveness and label clarity matter more than sample count.

Facial hair attributes (Mustache: precision=0.48, recall=0.87; Goatee: precision=0.55, recall=0.93) systematically over-predict due to visual confusion between adjacent fine-grained categories.

5.4 Ablation: Contribution of Each Design Decision

Design Decision	Approx. Macro F1	Note
Baseline: simple 2-layer CNN	0.320	Course starting point
+ DenseNet169 pretrained	~0.589	Single largest improvement
+ pos_weight (class imbalance)	~0.604	Critical for rare attributes
+ CLAHE preprocessing	~0.618	Local contrast enhancement
+ Per-class threshold tuning	~0.627	Direct F1 optimization on val set
+ SWA (final, test set)	0.7413	Generalization boost at no extra cost

Table 7. Approximate contribution of each design decision to final classifier performance.

5.5 cVAE Generation Results

Table 8 summarizes the six cVAE variants trained across two resolutions and three loss formulations. The 128×128 MSE + 1000·LPIPS + KL variant was the clear winner on macro F1 and smallest Δ while remaining competitive on FID.

Variant	Res.	Loss	FID ↓	Macro F1 (gen) ↑	Avg Δ
cVAE base	64	MSE + KL (warm-up 0→0.5)	104.27	0.4095	−0.3421
σ -VAE $\text{klw}=1$	64	σ -VAE	104.27	0.4095	−0.3421
σ -VAE $\text{klw}=100$	64	σ -VAE, high KL	100.71	0.2288	−0.5228
σ -VAE $\text{klw}=100$ + LPIPS	64	σ -VAE + perceptual	97.62	0.3239	−0.4278
Option 1 + LPIPS (winner)	128	MSE + 1000·LPIPS + KL	104.41	0.4300	−0.3217

Table 8. cVAE variant comparison. FID computed against 10k real CelebA images; Macro F1 and Δ on 10k generated images evaluated by Irene’s classifier.

The winning 128×128 model achieves macro F1=0.4300 on 10,000 generated images against a real-image ceiling of 0.7517, retaining approximately 57% of the classifier’s signal. Table 9 shows the best and worst-performing attributes in generation.

Attribute	F1 (generated)	F1 (real)	Avg Δ	Category
Male	0.9630	0.9864	−0.023	Global, salient
No_Beard	0.9539	0.9770	−0.023	Global, salient
Young	0.8749	0.9188	−0.044	Global, salient
Smiling	0.8723	0.9255	−0.053	Global, salient
Mouth_Slightly_Open	0.8508	0.9379	−0.087	Global, salient
Wearing_Earrings	0.0238	0.7890	−0.765	Small-area accessory
Wearing_Necklace	0.0623	0.5315	−0.469	Small-area accessory
Rosy_Cheeks	0.0679	0.6471	−0.579	Subtle skin property
Pale_Skin	0.0850	0.6068	−0.522	Subtle skin property
Double_Chin	0.1037	0.6401	−0.536	Subtle face shape

Table 9. Best and worst attributes in cVAE generation (10k generated images).

The pattern is interpretable: global, structurally salient attributes (Male, Smiling, Young) are generated near the classifier ceiling because they change the majority of facial

pixels, aligning reconstruction loss with conditioning. Small-area accessories (Wearing_Earrings, Wearing_Necklace) and subtle skin properties (Rosy_Cheeks, Pale_Skin) collapse almost completely because their pixel footprint is too small to drive the MSE + LPIPS objective.

5.6 Streamlit Demo Results

The Streamlit demo serves as an interactive platform that integrates generation, evaluation, and user interaction. To fully evaluate the system, we test two major modules: (1) attribute-controlled face generation with classifier evaluation, and (2) image upload with direct classification.

5.6.1 Generation and Evaluation Module

(a) Valid Attribute Generation

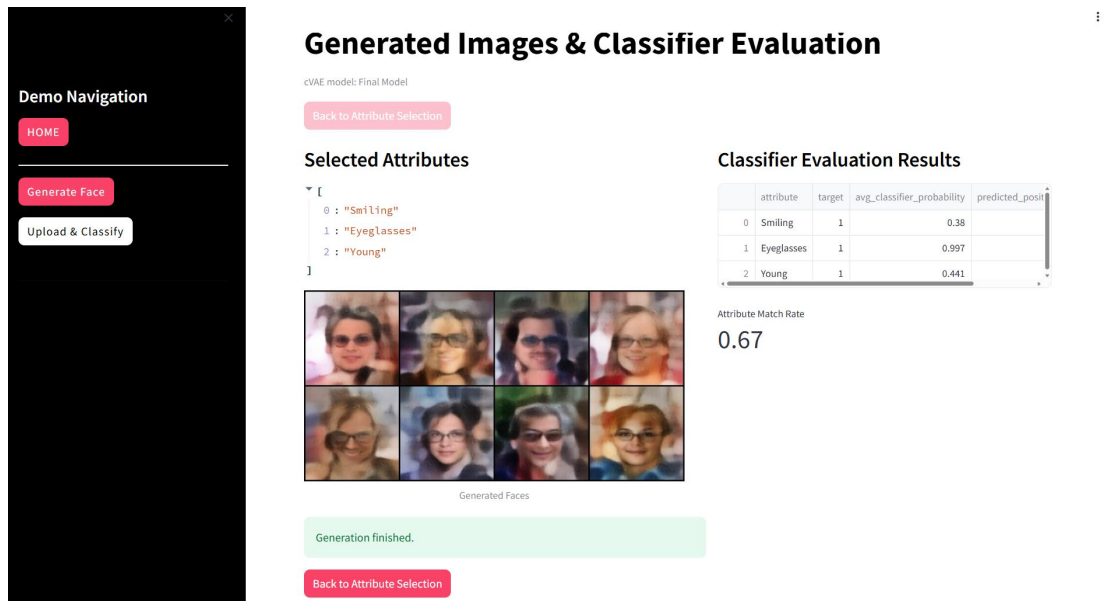


Figure 10. Generated images and classifier evaluation results for a valid attribute combination (Smiling, Eyeglasses, Young) using the updated cVAE model.

Figure 10 shows an example of generated images using a valid attribute combination (Smiling, Eyeglasses, Young). The generated faces are visually coherent, and key attributes such as eyeglasses and smiling are clearly observable by human inspection.

Attribute	Target	Avg Classifier Probability	Predicted Positive Rate	Match
Smiling	1	0.380	0.500	Yes

Eyeglasses	1	0.997	1.000	Yes
Young	1	0.441	0.375	No

Table 11. Classifier evaluation results for the valid attribute combination (Smiling, Eyeglasses, Young) using the updated cVAE model.

The classifier evaluation results are summarized in Table 11. Eyeglasses achieves a very high probability (0.997), indicating strong visual consistency. Smiling is also successfully detected, although with moderate confidence. However, the Young attribute is less stable, highlighting a gap between human perception and classifier-based evaluation.

This demonstrates that the system can successfully generate visually plausible images and provide quantitative feedback through the classifier.

(b) Conflict-Aware Generation

To evaluate system robustness, we tested both hard and soft conflicting attribute combinations.

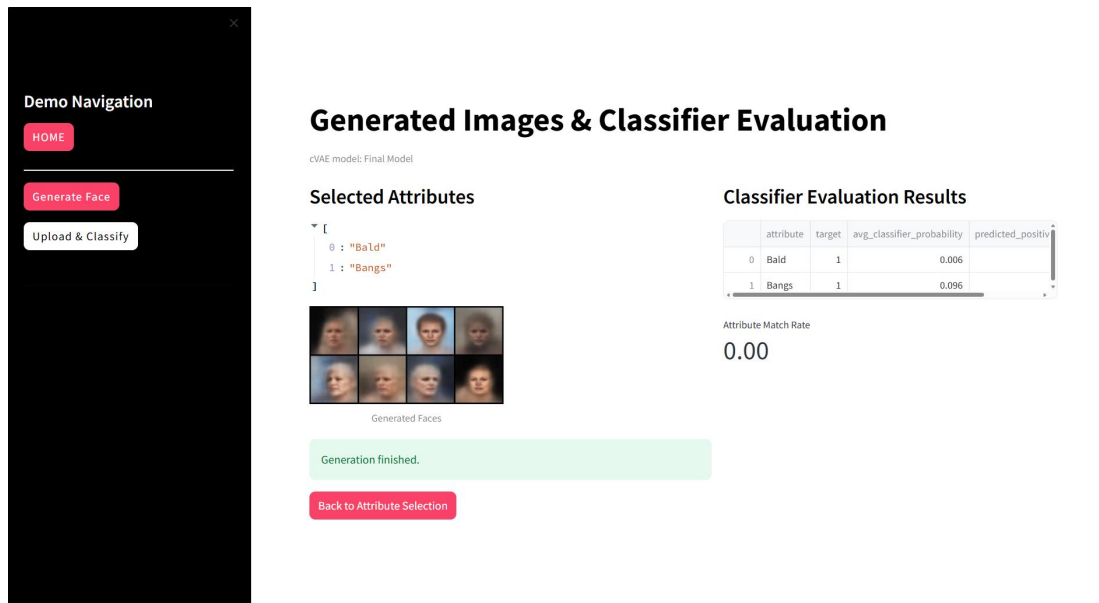


Figure 12. Generated images for a hard conflict combination (Bald + Bangs), showing unstable generation and failure to represent both attributes.

For hard conflicts (e.g., Bald + Bangs), Figure 12 shows that the generated images are unstable and fail to represent either attribute.

Attribute	Target	Avg Classifier Probability	Predicted Positive Rate	Match
Bald	1	0.096	0.000	No

Bangs	1	0.096	0.000	No
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Table 13. Classifier evaluation results for the hard conflict combination (Bald + Bangs), showing near-zero probabilities for both attributes.

The classifier results (Table 13) confirm this, with both attributes receiving near-zero probabilities.

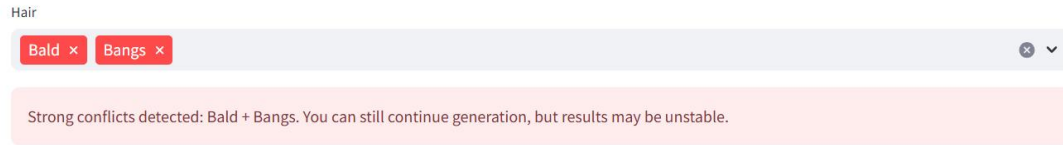


Figure 14. Interface display of hard conflict warning (Bald + Bangs), indicating that the system detects incompatible attribute combinations before generation.

Importantly, the system displays a real-time warning before generation, indicating that the conflict detection mechanism is functioning correctly.

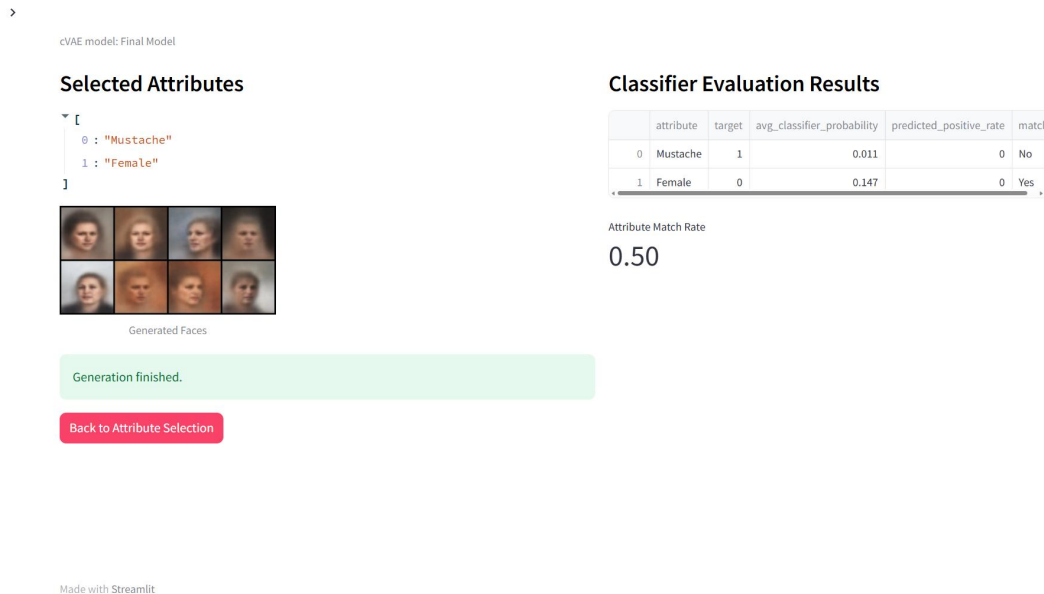


Figure 15. Generated images for a soft conflict combination (Female + Mustache), showing dominance of the primary attribute and suppression of the conflicting attribute.

For soft conflicts (e.g., Female + Mustache), Figure 15 shows that the generated images prioritize the dominant attribute (Female), while the rare attribute (Mustache) is largely ignored.

Attribute	Target	Avg Classifier Probability	Predicted Positive Rate	Match
Mustache	1	0.011	0.000	No
Female	0	0.147	0.000	Yes

Table 16. Classifier evaluation results for the soft conflict combination (Female + Mustache), showing imbalance between attributes.

The classifier results further confirm this imbalance.



Gender
Female

Age
Skip

Hair
Choose an option

Face Structure
Choose an option

Facial Hair
Mustache

Unusual combinations detected: Female + Mustache. These are rare in the dataset, but generation is still allowed.

Figure 17. Interface display of soft conflict warning (Female + Mustache), indicating an uncommon but allowed attribute combination.

A softer warning is displayed in the interface, indicating that the combination is uncommon but still allowed.

These experiments demonstrate that the conflict detection system not only improves usability but also provides insight into the limitations of the generative model under unrealistic conditions.

(c) Model Comparison (Old vs Updated cVAE)

A key feature of the demo is the ability to dynamically switch between different cVAE models. This enables direct comparison under identical input conditions.

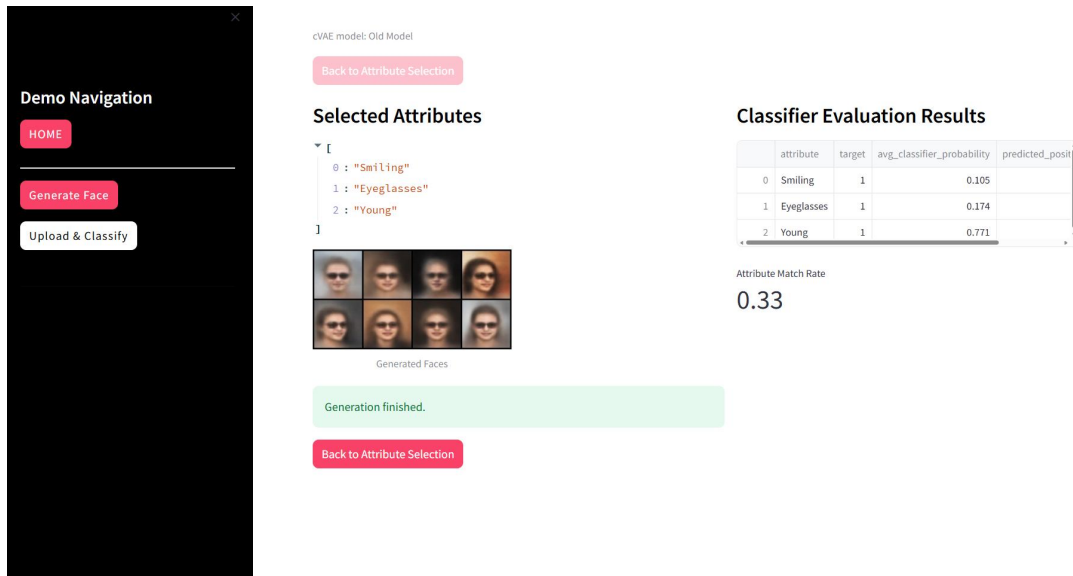


Figure 18. Generated images and classifier evaluation results for the original cVAE model under the attribute combination (Smiling, Eyeglasses, Young).

Attribute	Target	Avg Classifier Probability	Predicted Positive Rate	Match
Smiling	1	0.105	0.000	No
Eyeglasses	1	0.174	0.125	No
Young	1	0.771	1.000	Yes

Table 19. Classifier evaluation results for the original cVAE model, showing limited attribute alignment and lower match rate.

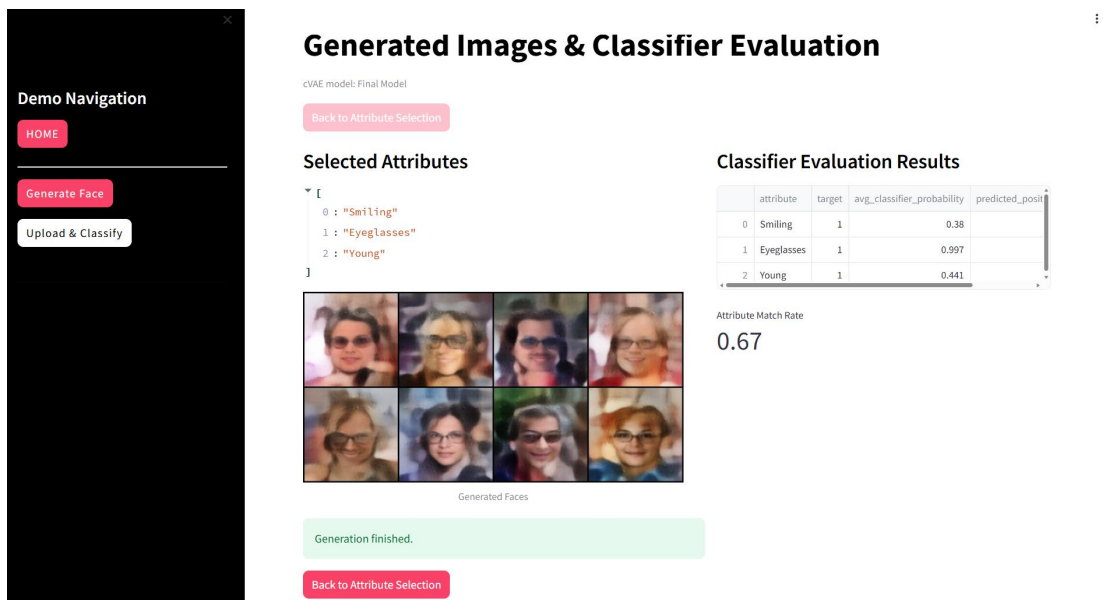


Figure 20. Generated images and classifier evaluation results for the updated cVAE model under the

same attribute combination.

Attribute	Target	Avg Classifier Probability	Predicted Positive Rate	Match
Smiling	1	0.380	0.500	Yes
Eyeglasses	1	0.997	1.000	Yes
Young	1	0.441	0.375	No

Table 21. Classifier evaluation results for the updated cVAE model, showing improved detection of visually salient attributes.

Using the same attribute combination (Smiling, Eyeglasses, Young), Figure 18 and Table 19 show the results for the original model, while Figure 20 and Table 21 show the results for the updated model.

The updated model achieves a higher match rate (0.67 vs 0.33) and significantly improves the generation of visually salient attributes such as eyeglasses and smiling. However, the Young attribute becomes less consistent, indicating that improvements are not uniform across all attributes.

This comparison highlights that different model architectures may prioritize different aspects of the data distribution. The ability to perform such comparisons interactively is a key advantage of the system.

5.6.2 Upload and Classification Module

(a) Real Face Image Classification

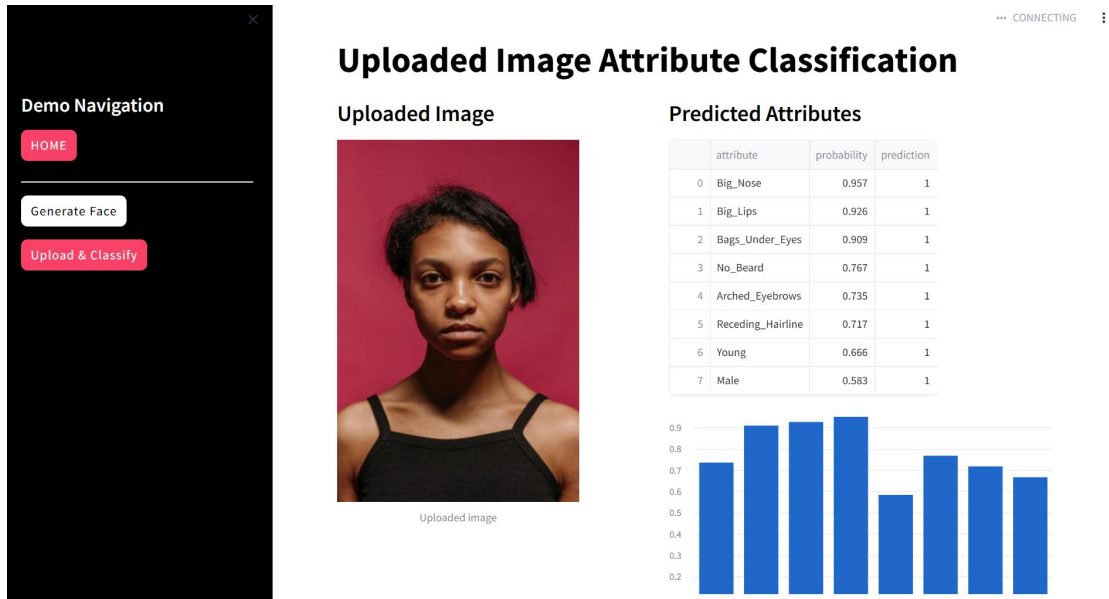


Figure 22. Uploaded real face image and corresponding classifier predictions, demonstrating generalization to external data.

The upload module allows users to input external images and directly apply the classifier. Figure 22 shows an example of a real face image uploaded from a public dataset. The classifier successfully identifies several attributes such as No_Beard and Young, demonstrating reasonable generalization to real-world inputs.

Attribute	Probability	Prediction
Big_Nose	0.957	1
Big_Lips	0.926	1
Bags_Under_Eyes	0.909	1
No_Beard	0.767	1
Arched_Eyebrows	0.735	1
Receding_Hairline	0.717	1
Young	0.666	1
Male	0.583	1

Table 23. Classifier predictions for the uploaded face image, including attribute probabilities and binary outputs.

The results are displayed both as a table and a probability bar chart, providing intuitive interpretation of classifier confidence.

(b) Non-Face Input Detection (Robustness)

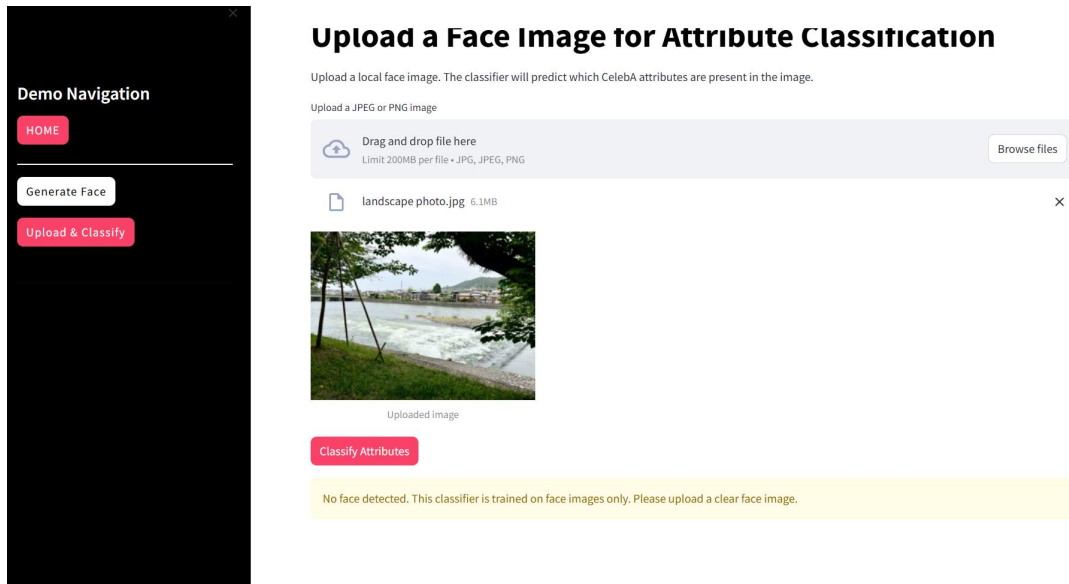


Figure 24. Interface response to a non-face input (landscape image), showing face detection warning and preventing invalid classification.

To ensure robustness, a face detection step is applied before classification. Figure 24 shows an example of a landscape image being uploaded. In this case, the system correctly detects that no face is present and displays a warning message instead of producing classifier outputs.

This prevents meaningless predictions and improves the reliability of the system when handling arbitrary user inputs.

5.6.3 Streamlit demo summary

Overall, the Streamlit demo demonstrates not only the functionality of the underlying models, but also the importance of system-level design. Features such as conflict detection, model selection, and input validation transform the demo into a flexible experimental platform for analyzing generative model behavior under various conditions.

6. Summary and Conclusions

6.1 Summary

This project developed a three-component deep learning pipeline for attribute-controlled face generation and evaluation. The DenseNet169 + BiLSTM classifier achieves macro F1=0.7413 on 40 attributes, a 2.3× improvement over the simple CNN baseline. The 128×128 cVAE achieves macro F1=0.4300 on 10,000 generated images (retaining 57% of the real-image signal) with FID=104.41. The Streamlit demo integrates both models with hard/soft conflict detection for live user exploration, and further demonstrates how model integration, conflict-aware interaction design, and real-time evaluation can provide insights into both model capabilities and limitations.

6.2 Key Findings

- Visual distinctiveness predicts attribute F1 better than sample count. Eyeglasses (1,289 positives, F1=0.978) far outperforms Oval_Face (5,901 positives, F1=0.565) because glasses are visually unambiguous.
- Conditioning fidelity is highly attribute-dependent and is not captured by single-number metrics like FID. Global salient attributes (Male, Young, Smiling) are generated near the classifier ceiling; small-area accessories (earrings, necklaces) and subtle skin properties collapse almost entirely.
- The root cause of small-area failure is the reconstruction objective: MSE + LPIPS losses are dominated by the majority of facial pixels, and the absence of earrings or subtle cheek color is not penalized enough to make the decoder synthesize them reliably.
- KL warm-up is essential for cVAE training. Without it, posterior collapse occurs in the first few epochs and the model ignores z entirely, producing nearly identical images regardless of the latent sample.
- SWA consistently improves classifier generalization (+0.012 val F1) at no additional training cost and is recommended as a default for multi-label classification.

6.3 Future Work

- Attribute-weighted reconstruction loss: upweight pixels in regions associated with rare accessories (ear, neck areas) to directly address the small-area failure mode.
- Auxiliary classifier-guided loss: backpropagate BCE from the classifier through the cVAE during training to close the generation-evaluation loop.
- Higher resolution (256×256): gives small accessories more pixels to land on, at the cost of substantially more compute.

- Spatial attention in the classifier: replace BiLSTM with attention over the feature map, dynamically focusing on the most relevant facial region per attribute.
- Dedicated facial hair submodel: the Mustache/Goatee/Sideburns confusion motivates a fine-grained multi-class model for this sub-category.

7. References

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8. Appendix: Code Structure

All code is available in the GitHub repository: <https://github.com/zcardell-gwu/Final-Project-Group3>

File	Developer	Description
train script.py	Irene Xia	Multi-label classifier: DenseNet169+BiLSTM, CLAHE caching, SWA, threshold tuning, full training loop
cVAE_train_option1_lpips.py	Zachary Cardell	Winning cVAE: 128×128 architecture, MSE+LPIPS+KL loss, KL warm-up, training loop
cVAE_train_revised.py	Zachary Cardell	Revised training script with final model saving logic
evaluate_generated_full.py	Zachary Cardell	10k evaluation pipeline: generation, classifier inference, per-attribute F1, CLIP cross-check
download_data.py	Zachary Cardell	CelebA download/extract, preprocessing, attribute parsing
streamlit_demo.py	Jiatong Peng	Streamlit app: attribute selection, conflict detection, generation, evaluation, image upload

Table 25. Code files in the repository.